EWB CLOUD LABS STUDENT PROJECT

AWS VPC Peering — Project Report

Draft report template — update with your actual AWS resource IDs, test results, and console screenshots before submission.

Student Name	_____
USN / Roll No.	_____
Class / Batch	_____
Faculty Name	_____
AWS Region	Asia Pacific (Sydney) — ap-southeast-2 (confirm in console)
Submission Date	_____

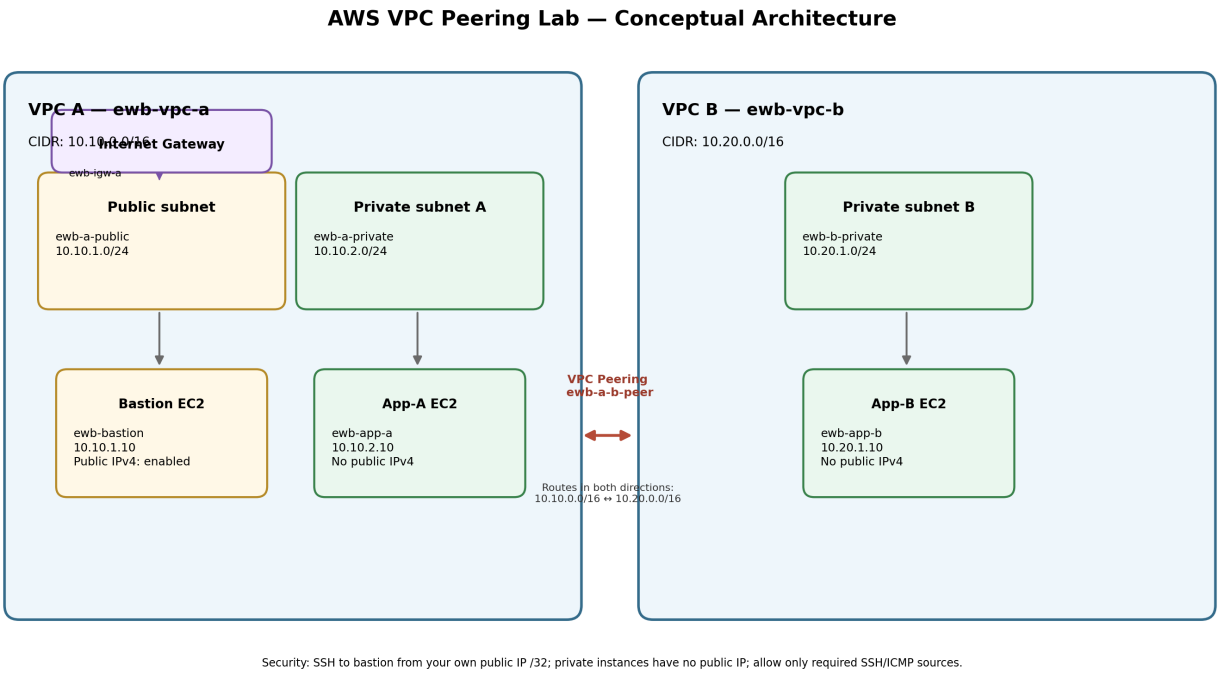
Evidence integrity note: This document contains a designed architecture diagram and evidence placeholders. It does not contain real AWS console screenshots or claim that the EC2 instances and connectivity tests succeeded. Add your own screenshots after the resources are created and tested.

1. Aim

To create two Amazon Virtual Private Clouds (VPCs), connect them using VPC Peering, configure routes and security groups, and verify private communication between EC2 instances while keeping the application instances without public IPv4 addresses.

2. Architecture Diagram

The following diagram represents the intended topology from the project workbook. Confirm the actual resource names and addresses in your AWS console before final submission.



3. Planned AWS Resource Inventory

Resource	Name	Configuration / CIDR
VPC A	ewb-vpc-a	10.10.0.0/16
Public subnet A	ewb-a-public	10.10.1.0/24
Private subnet A	ewb-a-private	10.10.2.0/24
VPC B	ewb-vpc-b	10.20.0.0/16
Private subnet B	ewb-b-private	10.20.1.0/24
Internet Gateway	ewb-igw-a	Attached to VPC A
VPC peering	ewb-a-b-peer	VPC A ↔ VPC B; status should be Active
Bastion EC2	ewb-bastion	10.10.1.10; public IPv4 enabled
App-A EC2	ewb-app-a	10.10.2.10; no public IPv4
App-B EC2	ewb-app-b	10.20.1.10; no public IPv4
Public route table	ewb-rt-a-public	Local VPC A; 0.0.0.0/0 → IGW; 10.20.0.0/16 → peering
Private route table A	ewb-rt-a-private	Local VPC A; 10.20.0.0/16 → peering
Private route table B	ewb-rt-b-private	Local VPC B; 10.10.0.0/16 → peering

4. Routing and Security Design

• **Public route table:** retain the local route for 10.10.0.0/16, send 0.0.0.0/0 to the Internet Gateway, and send 10.20.0.0/16 to the peering connection.

• **Private route table A:** route 10.20.0.0/16 to the peering connection.

• **Private route table B:** route 10.10.0.0/16 to the peering connection.

• **Bastion security group:** inbound SSH TCP 22 only from the student's current public IPv4 address as /32 (AWS “My IP” option).

• **App-A and App-B security groups:** SSH TCP 22 only from the bastion private IP 10.10.1.10/32; ICMP Echo Request only from the peer application instance IP, as required by the lab.

• **Outbound rules:** use the workbook's default outbound allowance unless your instructor specifies otherwise. Do not expose SSH to 0.0.0.0/0.

5. Implementation Procedure

1. Create VPC A (10.10.0.0/16), its public and private subnets, and attach the Internet Gateway to VPC A.

2. Create the public route table and associate it with the public subnet; add the Internet Gateway default route.

3. Create VPC B (10.20.0.0/16), its private subnet, and its private route table.

4. Create the VPC peering connection between VPC A and VPC B, then accept it if prompted.

5. Add peering routes to the public route table and both private route tables.

6. Create the security groups with least-privilege SSH and ICMP rules.

7. Create or select the EC2 key pair. Store the private key securely on the laptop; never upload it to GitHub or include it in this report.

8. Launch the bastion in the public subnet and the two application instances in the private subnets. Use the instructor-approved instance type and check estimated costs before launching.

9. Verify that App-A and App-B have no public IPv4 addresses.

10. Run and record the baseline and post-peering connectivity tests. Capture genuine AWS console and terminal screenshots.

6. Verification Results — Fill in After Testing

Test	Expected result	Actual result / evidence
Peering status	Active	_____
Baseline ping before routes/peering	Should fail before peering path is configured	_____
App-A → App-B ping after routes and ICMP rules	Replies received	_____
App-B → App-A ping after routes and ICMP rules	Replies received	_____
SSH to App-B through bastion	Successful, if configured as workbook requires	_____
Controlled ICMP rule removal	Ping fails while rule is removed	_____
Restore ICMP rule	Ping succeeds again	_____
Private instance public IPv4	None for App-A and App-B	_____

7. AWS Screenshot Evidence — Insert Actual Screenshots

Replace each placeholder with a screenshot captured from your own AWS console or terminal. Keep the AWS Region visible where practical, and do not show private keys, passwords, tokens, or other secrets.

Figure 1 — VPCs and CIDR blocks
EC2/VPC console showing ewb-vpc-a and ewb-vpc-b with their CIDRs.
[Insert actual screenshot here]

Figure 2 — Subnets
Subnets page showing all three subnets, their VPCs, and CIDR blocks.
[Insert actual screenshot here]

Figure 3 — Route tables
Routes and subnet associations for all three route tables, including peering targets.
[Insert actual screenshot here]

Figure 4 — VPC peering
Peering connection details with status Active and both VPC IDs.
[Insert actual screenshot here]

Figure 5 — Security groups
Inbound rules for bastion, App-A, and App-B; hide any unrelated sensitive information.
[Insert actual screenshot here]

Figure 6 — EC2 instances
Instance list/details showing names, states, private IPs, and no public IPv4 for App-A/App-B.
[Insert actual screenshot here]

Figure 7 — Connectivity test
Terminal output showing real ping results in both directions after routes and rules are configured.
[Insert actual screenshot here]

Figure 8 — SSH test
Terminal evidence showing SSH access to App-B through the bastion; do not expose key contents.
[Insert actual screenshot here]

Figure 9 — Failure and recovery
Evidence of the temporary ICMP rule test and successful recovery after restoring the rule.
[Insert actual screenshot here]

Figure 10 — Cleanup
Console evidence that billable test resources were stopped/terminated and resources cleaned up, if required.
[Insert actual screenshot here]

8. Results and Conclusion

The intended outcome is private communication between instances in two separate VPCs through VPC Peering, controlled by route tables and security groups. Complete this section only after testing: state which tests passed, which failed, how any errors were resolved, and whether the private instances remained without public IPv4 addresses. Do not claim success unless your actual test evidence supports it.

9. Safety, Security, and Cost Notes

• Do not set inbound SSH source to 0.0.0.0/0. Use your own public IPv4 /32 for the bastion.

• Never upload the .pem private key to GitHub, email, or this report.

• Check AWS pricing and your account's billing/credits before launching instances. EC2, EBS storage, and public IPv4 addresses may incur charges.

• After collecting evidence, terminate test EC2 instances and remove unused resources if your instructor does not require them to remain running. Verify cleanup in the console.

• Use only real screenshots from your own account. The diagram in this report is conceptual and is not proof of deployment.